Homework 3 Solutions: Linear Algebra

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Practice Problems: Work together in class (The matrix solutions need to be done via the inverse method)

1. Consider the following system of equations:

$$\begin{aligned} 2x_1 + x_2 &= 10 \\ x_1 + 3x_2 &= 15. \end{aligned}$$

(a) The system can be written in matrix form as

$$\mathbf{Ax} = \mathbf{b},$$

where

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \end{bmatrix}.$$

(b) The coefficient matrix is

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}.$$

The vector of unknowns is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

The right-hand-side vector is

$$\mathbf{b} = \begin{bmatrix} 10 \\ 15 \end{bmatrix}.$$

(c) From the first equation,

$$x_2 = 10 - 2x_1.$$

Substitute this into the second equation:

$$x_1 + 3(10 - 2x_1) = 15.$$

Therefore,

$$x_1 + 30 - 6x_1 = 15,$$

so

$$-5x_1 = -15.$$

Hence,

$$x_1 = 3.$$

Then

$$x_2 = 10 - 2(3) = 4.$$

Therefore, the solution is

$$x_1 = 3, \qquad x_2 = 4.$$

(d) If x_1 and x_2 are quantities of two goods, then the solution says that choosing 3 units of good 1 and 4 units of good 2 satisfies both linear requirements exactly.

2. Let

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

(a)

$$A + B = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} + \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 1 & 5 \end{bmatrix}.$$

(b)

$$A - C = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

(c)

$$3A = 3\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 3 & 9 \end{bmatrix}.$$

(d)

$$2B - C = 2\begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

First,

$$2B = \begin{bmatrix} 10 & -2 \\ 0 & 4 \end{bmatrix}.$$

Therefore,

$$2B - C = \begin{bmatrix} 10 & -2 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 9 & -4 \\ -3 & 0 \end{bmatrix}.$$

3. Suppose

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

records input requirements, where rows represent goods and columns represent inputs. Let

$$\mathbf{q} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}.$$

(a) First compute

$$A' = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}.$$

Then

$$A'\mathbf{q} = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 2(10) + 4(5) \\ 1(10) + 3(5) \end{bmatrix} = \begin{bmatrix} 40 \\ 25 \end{bmatrix}.$$

(b) The first entry, 40, is the total amount of labor required. The second entry, 25, is the total amount of capital required.

(c) We use A' because the rows of A represent goods and the columns represent inputs. Multiplying by A' aggregates input requirements across the quantities of goods produced.

4. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}.$$

(a)

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1(2) + 2(1) & 1(0) + 2(3) \\ 3(2) + 4(1) & 3(0) + 4(3) \end{bmatrix}.$$

Therefore,

$$AB = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix}.$$

(b)

$$BA = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2(1)+0(3) & 2(2)+0(4) \\ 1(1)+3(3) & 1(2)+3(4) \end{bmatrix}.$$

Therefore,

$$BA = \begin{bmatrix} 2 & 4 \\ 10 & 14 \end{bmatrix}.$$

(c) No,

$$AB \neq BA.$$

(d) This example shows that matrix multiplication is generally not commutative. In general,

$$AB \neq BA.$$

5. For each matrix below, determine whether it is an identity matrix, a null matrix, a symmetric matrix, or none of these. Comment on the linear independence and determinant in each case.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & 5 \\ 5 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

(a) The matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is the identity matrix. It is also symmetric because

$$A' = A.$$

Its determinant is

$$|A| = 1(1) - 0(0) = 1.$$

Since the determinant is nonzero, its columns are linearly independent.

(b) The matrix

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

is the null matrix. It is also symmetric. Its determinant is

$$|B| = 0(0) - 0(0) = 0.$$

Since the determinant is zero, its columns are linearly dependent.

(c) The matrix

$$C = \begin{bmatrix} 2 & 5 \\ 5 & 3 \end{bmatrix}$$

is symmetric because

$$C' = C.$$

It is not the identity matrix and it is not the null matrix. Its determinant is

$$|C| = 2(3) - 5(5) = 6 - 25 = -19.$$

Since the determinant is nonzero, its columns are linearly independent.

(d) The matrix

$$D = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

is not the identity matrix, not the null matrix, and not symmetric. Its determinant is

$$|D| = 1(1) - 2(0) = 1.$$

Since the determinant is nonzero, its columns are linearly independent.

Homework Problems: Submit individually

6. Consider the following simple market model:

$$Q_d = 40 - 2P$$
$$Q_s = 10 + P.$$

In equilibrium, $Q_d = Q_s = Q$.

(a) Since $Q = Q_d$ and $Q = Q_s$, we have

$$Q = 40 - 2P$$

and

$$Q = 10 + P.$$

Rearranging,

$$Q + 2P = 40$$

and

$$Q - P = 10.$$

(b) In matrix form,

$$\begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} Q \\ P \end{bmatrix} = \begin{bmatrix} 40 \\ 10 \end{bmatrix}.$$

(c) From the system,

$$Q + 2P = 40$$

and

$$Q - P = 10.$$

Subtract the second equation from the first:

$$(Q + 2P) - (Q - P) = 40 - 10.$$

Thus,

$$3P = 30,$$

so

$$P^* = 10.$$

Then

$$Q^* = 10 + P^* = 10 + 10 = 20.$$

(d) The equilibrium price is

$$P^* = 10,$$

and the equilibrium quantity is

$$Q^* = 20.$$

At this price, quantity demanded equals quantity supplied.

7. Let

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

(a) The determinant is

$$|A| = 4(3) - 2(1) = 12 - 2 = 10.$$

(b) Since

$$|A| = 10 \neq 0,$$

the inverse exists.

(c) For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Therefore,

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix}.$$

(d) We want to solve

$$A\mathbf{x} = \begin{bmatrix} 14 \\ 8 \end{bmatrix}.$$

Therefore,

$$\mathbf{x} = A^{-1} \begin{bmatrix} 14 \\ 8 \end{bmatrix}.$$

Substituting,

$$\mathbf{x} = \frac{1}{10} \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 14 \\ 8 \end{bmatrix}.$$

Multiplying,

$$\mathbf{x} = \frac{1}{10} \begin{bmatrix} 3(14) - 2(8) \\ -1(14) + 4(8) \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 42 - 16 \\ -14 + 32 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 26 \\ 18 \end{bmatrix}.$$

Thus,

$$\mathbf{x} = \begin{bmatrix} 13/5 \\ 9/5 \end{bmatrix}.$$

8. Evaluate the determinants of the following matrices. Then state whether each matrix is singular or nonsingular.

(a)

$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \end{bmatrix}.$$

The determinant is

$$|A| = 3(2) - 1(6) = 6 - 6 = 0.$$

Therefore, A is singular.

(b)

$$B = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}.$$

The determinant is

$$|B| = 2(4) - 5(1) = 8 - 5 = 3.$$

Since

$$|B| = 3 \neq 0,$$

the matrix B is nonsingular.

(c)

$$C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 0 & 1 \end{bmatrix}.$$

Expanding along the first row,

$$|C| = 1 \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} + 0 \begin{vmatrix} 0 & 3 \\ 2 & 0 \end{vmatrix}.$$

Therefore,

$$|C| = 1(3 \cdot 1 - 1 \cdot 0) - 2(0 \cdot 1 - 1 \cdot 2) = 3 - 2(-2) = 7.$$

Since

$$|C| = 7 \neq 0,$$

the matrix C is nonsingular.

9. Consider the input-output system

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 18 \\ 21 \end{bmatrix}.$$

(a) The coefficient matrix is

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its determinant is

$$|A| = 2(2) - 1(1) = 4 - 1 = 3.$$

By Cramer's Rule,

$$x_1 = \frac{|A_1|}{|A|},$$

where A_1 replaces the first column of A with the right-hand-side vector:

$$A_1 = \begin{bmatrix} 18 & 1 \\ 21 & 2 \end{bmatrix}.$$

Thus,

$$|A_1| = 18(2) - 1(21) = 36 - 21 = 15.$$

Therefore,

$$x_1 = \frac{15}{3} = 5.$$

Similarly,

$$x_2 = \frac{|A_2|}{|A|},$$

where

$$A_2 = \begin{bmatrix} 2 & 18 \\ 1 & 21 \end{bmatrix}.$$

Then

$$|A_2| = 2(21) - 18(1) = 42 - 18 = 24.$$

Therefore,

$$x_2 = \frac{24}{3} = 8.$$

(b) Substitute $x_1 = 5$ and $x_2 = 8$ into the original system:

$$2(5) + 8 = 10 + 8 = 18,$$

and

$$5 + 2(8) = 5 + 16 = 21.$$

Both equations are satisfied.

(c) Yes, the coefficient matrix is nonsingular because

$$|A| = 3 \neq 0.$$

(d) Nonsingularity implies that the system has a unique solution.

10. **Slightly harder problem.** Suppose a consumer's total spending on two goods must satisfy two linear restrictions:

$$\begin{aligned} 2x_1 + 4x_2 &= 20 \\ x_1 + 2x_2 &= 10. \end{aligned}$$

(a) The system in matrix form is

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \end{bmatrix}.$$

(b) The determinant of the coefficient matrix is

$$\begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix} = 2(2) - 4(1) = 4 - 4 = 0.$$

(c) No, the two equations are not linearly independent. The first equation is exactly two times the second equation:

$$2(x_1 + 2x_2 = 10) \quad \Rightarrow \quad 2x_1 + 4x_2 = 20.$$

(d) No, the system does not have a unique solution. Since the two equations are the same restriction, there are infinitely many solutions satisfying

$$x_1 + 2x_2 = 10.$$

(e) Economically, the two restrictions do not provide two independent pieces of information because they describe the same budget or spending restriction. The first equation is just a rescaled version of the second.

(f) One different second equation that would make the coefficient matrix nonsingular is

$$x_1 + x_2 = 10.$$

Then the coefficient matrix would be

$$\begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix}.$$

Its determinant is

$$2(1) - 4(1) = -2 \neq 0,$$

so the matrix would be nonsingular.

11. **Based on class discussion.** Suppose

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

(a) A general linear combination is

$$a\mathbf{v}_1 + b\mathbf{v}_2 = a \begin{bmatrix} 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Therefore,

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \begin{bmatrix} 2a - b \\ a + 2b \end{bmatrix}.$$

(b) We want

$$\begin{bmatrix} 2a - b \\ a + 2b \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

This gives the system

$$2a - b = 3, \qquad a + 2b = 4.$$

From the first equation,

$$b = 2a - 3.$$

Substitute into the second equation:

$$a + 2(2a - 3) = 4.$$

Thus,

$$a + 4a - 6 = 4,$$

so

$$5a = 10.$$

Therefore,

$$a = 2.$$

Then

$$b = 2(2) - 3 = 1.$$

So yes, the activities can generate

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

using

$$a = 2, \qquad b = 1.$$

(c) We want

$$\begin{bmatrix} 2a - b \\ a + 2b \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}.$$

This gives

$$2a - b = 1, \qquad a + 2b = 7.$$

From the first equation,

$$b = 2a - 1.$$

Substitute into the second equation:

$$a + 2(2a - 1) = 7.$$

Therefore,

$$a + 4a - 2 = 7,$$

so

$$5a = 9.$$

Hence,

$$a = \frac{9}{5}.$$

Then

$$b = 2\left(\frac{9}{5}\right) - 1 = \frac{18}{5} - \frac{5}{5} = \frac{13}{5}.$$

So yes, the activities can generate

$$\begin{bmatrix} 1 \\ 7 \end{bmatrix},$$

using

$$a = \frac{9}{5}, \qquad b = \frac{13}{5}.$$

(d) The matrix with columns $\mathbf{v}_1$ and $\mathbf{v}_2$ is

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Its determinant is

$$|A| = 2(2) - (-1)(1) = 4 + 1 = 5.$$

Since

$$|A| = 5 \neq 0,$$

the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.

(e) Yes, $\mathbf{v}_1$ and $\mathbf{v}_2$ span $\mathbb{R}^2$. In $\mathbb{R}^2$, two linearly independent vectors span the whole space.

(f) Economically, if these two activity vectors span $\mathbb{R}^2$, then combinations of the two activities can generate any desired change in the two-dimensional output vector. In other words, the activities are flexible enough to move the economy in any direction in this two-good space.